

[Question 1] (a) In the Multi-armed Bandits problem, what is the cardinality of the state space $\mathcal{S}$?

(b) In Reinforcement Learning (RL) approaches to solve the multi-armed bandits, how do we estimate the reward distributions R from the experience collecting real world data?

(c) List all of the approaches we have considered to deal with the exploration-exploitation trade-off in the multi-armed bandits problem. What are the hyperparameters of such approaches? Show such hyperparameters by using equations if helpful and illustrate what will happen if we make such hyperparameters bigger.

Note: All answers to Question 1, including its subpoints, must be written exclusively on this side of the sheet. Answers written on other sheets or on the draft page will not be considered.

[Question 2] (a) What are the main motivations to use Dynamic Programming (DP) approaches?
(b) What are the main characteristic of a problem that make it feasible to be tackled with DP?
(c) Provide the pseudo-code for Policy Iteration ('in place' update version). Which mathematical tools are you using in this algorithm? **Note: All answers to Question 2, including its subpoints, must be written exclusively on this side of the sheet. Answers written on other sheets or on the draft page will not be considered.**

[Question 3] (a) In the full RL problem, what are the motivations to use off-policy approaches? What requirements are needed in order to employ off-policy approaches in the full RL setting?

(b) Consider a maze, where agents have the possibility to turn left or right during their exploration; an agent 'Alice' is exploring the maze, by choosing to turn left or right in the maze with uniform probability. Agent 'Alice' collect deterministically a -1 reward for each time he chooses to turn left or right, and a +100 reward for each time that he is exiting the maze (at that point the episode is terminated). Agent 'Bob' has instead a policy that will make him choose to go right with 100% probability. Agent 'Bob' collects deterministically 0 reward for each choice (left or right) that he makes and a +100 reward when he is exiting the maze (terminal state). We have available several trajectories collected by Agent 'Alice': can we use such trajectories to learn how good the policy of Agent 'Bob' is?

Note: All answers to Question 3, including its subpoints, must be written exclusively on this side of the sheet. Answers written on other sheets or on the draft page will not be considered.

[Question 4] (a) Report the pseudo-code and backup diagrams for SARSA and Expected SARSA.

(b) What are the main differences of the two approaches?

(c) Monte Carlo and TD-learning approaches converge to two different solutions, which ones?

Note: All answers to Question 4, including its subpoints, must be written exclusively on this side of the sheet. Answers written on other sheets or on the draft page will not be considered.